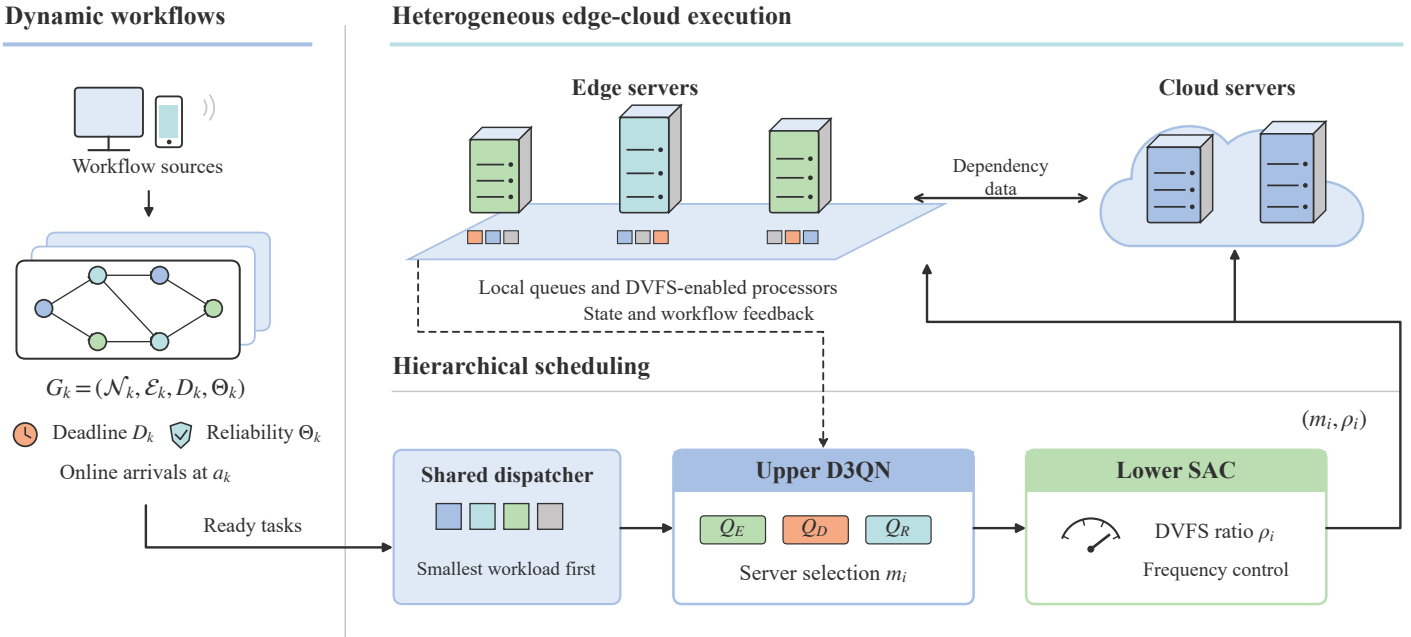

Dynamic workflows
Heterogeneous edge-cloud execution
Workflow sources
$G_k = (\mathcal{N}_k, \mathcal{E}_k, D_k, \Theta_k)$
Deadline D_k
Reliability Θ_k
Online arrivals at a_k
Ready tasks
Edge servers
Cloud servers
Dependency data
Local queues and DVFS-enabled processors
State and workflow feedback
Hierarchical scheduling
(m_i, ρ_i)
Shared dispatcher
Smallest workload first
Upper D3QN
Q_E Q_D Q_R
Server selection m_i
Lower SAC
DVFS ratio ρ_i
Frequency control